

Fine-Tuning Microsoft Phi-2 for Medical Data

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1 Introduction

This document explains the process of fine-tuning the Microsoft Phi-2 language model on a medical dataset, preparing it for question–answer style interactions. We describe dataset preparation, training with LoRA adapters, inference, and show how the model improves after fine-tuning.

2 Dataset Preparation

The dataset is provided in CSV format with two columns: `text` (the medical question or passage) and `label` (the corresponding answer). Each row is reformatted into an instruction-answer style:

```
Text:
<question>

Label:
<answer>
```

Example

```
Text:
What are the treatments for Acanthoma?

Label:
Surgical removal, cryotherapy, or observation
```

This formatting ensures the model learns to map medical questions to correct answers.

3 Fine-Tuning Phi-2

The fine-tuning function performs the following steps:

1. **Quantization:** Load the base Phi-2 model in 4-bit precision to save memory.
2. **LoRA Adaptation:** Apply Low-Rank Adaptation to query and value projections for efficient fine-tuning.
3. **Tokenization:** Convert text to token IDs, with a maximum sequence length of 1024 tokens.

4. **Trainer Setup:** Use HuggingFace's `SFTTrainer` with:

- Batch size: 4
- Gradient accumulation steps: 2
- Learning rate: 5×10^{-5}
- Epochs: 3
- Cosine scheduler with 10% warmup
- Mixed precision (bf16 or fp16)

5. **Save Model:** Merge LoRA adapters into the base model and save along with the tokenizer.

4 Assistant Wrapper

A Python class `CodeGenAssistant` is used for inference. It:

- Loads the fine-tuned model and tokenizer.
- Provides an `instruct()` method to format questions and generate answers.

```
assistant = CodeGenAssistant("/content/.../phi2-finetuned")
print(assistant.instruct("What are the treatments for Acanthoma?"))
```

5 Testing Before Fine-Tuning

For comparison, the base Phi-2 model can be tested on the same question:

```
response = test_phi2_before_finetune(
    "What are the treatments for Acanthoma?"
)
print(response)
```

Before fine-tuning, the responses are often generic or incomplete.

6 Illustrative Input–Output Comparison

Below is an example of how the model responds to a medical question before and after fine-tuning:

Question: What are the treatments for Acanthoma?

Before Fine-Tuning: It depends on context. Treatment may vary.

After Fine-Tuning: Surgical excision, cryotherapy, or observation, depending on severity.

7 Summary

In summary, the notebook and pipeline:

- Load a medical Q–A dataset and reformat examples for instruction-style training.
- Fine-tune Microsoft Phi-2 using LoRA adapters and 4-bit quantization for memory efficiency.
- Provide a simple assistant wrapper for inference.
- Demonstrate clear improvement in domain-specific answers post fine-tuning.